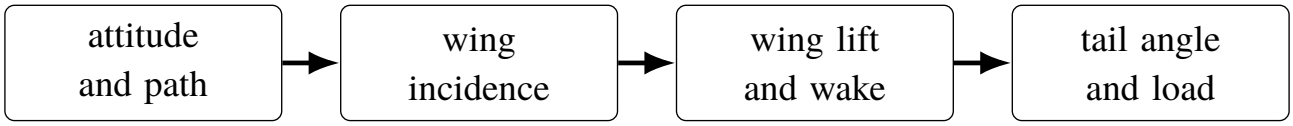
Week 2: Lift platforms angles and fundamental aerodynamics

The starting point. Week 1 introduced the aircraft's axes and motion. The next question is how that motion sets the angles at which the wing and tail meet the air, and how those angles produce aerodynamic loads.

> **The question for this lecture:** if the aircraft's attitude, flight path and control settings are known, what lift will its wing and tail produce?



Follow the sequence from aircraft geometry to local airflow and aerodynamic loading. The tail sits in air already disturbed by the wing.

The route through the lecture

1. Use aircraft attitude, flight path and wing setting to find chord incidence, then locate its zero-lift reference.
2. Recap aerofoil geometry, lift, drag and the effect of camber.
3. Extend the section model to a finite wing and its wake.
4. Construct tail incidence and add elevator and tab effects.

Today the aim is to predict the loads produced by prescribed angles and controls. Week 3 will use force and moment balance to find the loads needed for equilibrium.

Three questions guide the lecture

This lecture develops three questions:

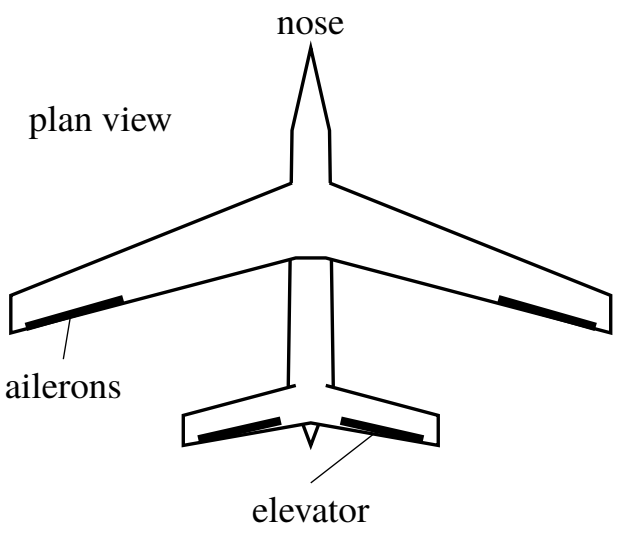
1. **What angle does the wing meet the air at?**
   Use aircraft attitude, flight path and wing setting, then locate the zero-lift reference.

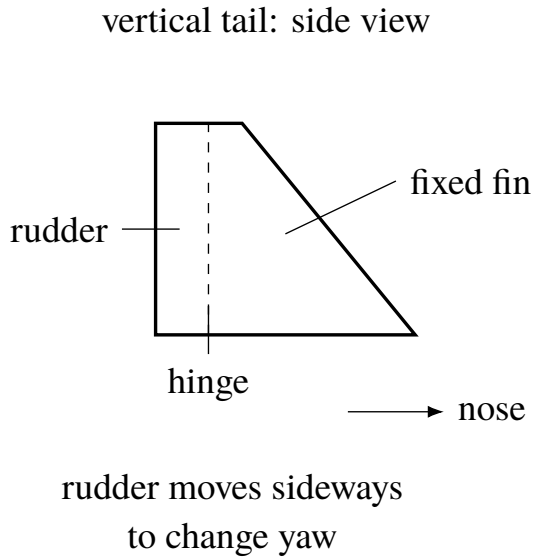
2. **How does that angle produce aerodynamic loading?**
   Use the aerofoil response and account for the finite wing's induced flow.

3. **What happens at the tail?**
   Find its local incidence, include the controls, and distinguish tail force from elevator hinge torque.

Working assumptions. A rigid, symmetric aircraft in predominantly low-speed flow. Linear lift laws apply in the attached-flow range.

The surfaces and their controls





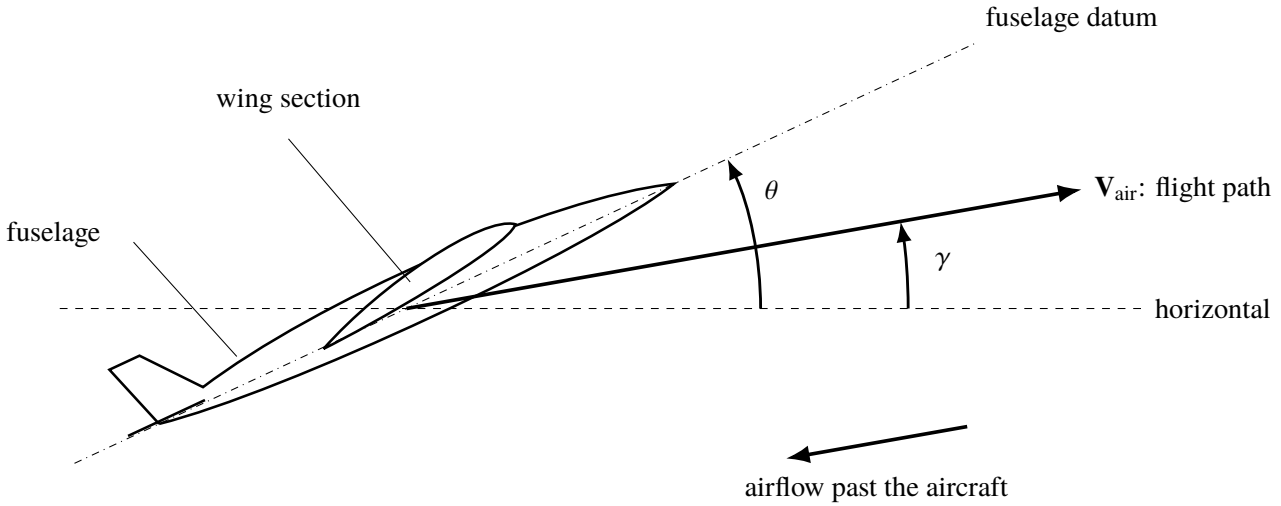
Surface	Control	Principal rotational effect
Wing	ailerons	roll
Horizontal tail	elevator	pitch
Vertical tail	rudder	yaw

The elevator changes the horizontal tail's aerodynamic loading. A small tab on the elevator also changes its hinge moment.

This week develops the wing and horizontal tail. Lateral-directional behaviour returns in Week 6.

1 / AIRCRAFT ANGLES

Separate aircraft attitude from flight path



Side view with a fuselage and a wing section. The datum is fixed in the fuselage; $\mathbf{V}_{\text{air}}$ is the aircraft velocity relative to the air.

θ	pitch attitude: fuselage datum above the horizontal
γ	flight-path angle: air-relative velocity above the horizontal

The fuselage datum is $\theta - \gamma$ above the forward air-relative velocity direction.

The airflow past the aircraft travels in the opposite direction to $\mathbf{V}_{\text{air}}$. Using forward-pointing reference directions allows comparison of the small incidence angles.

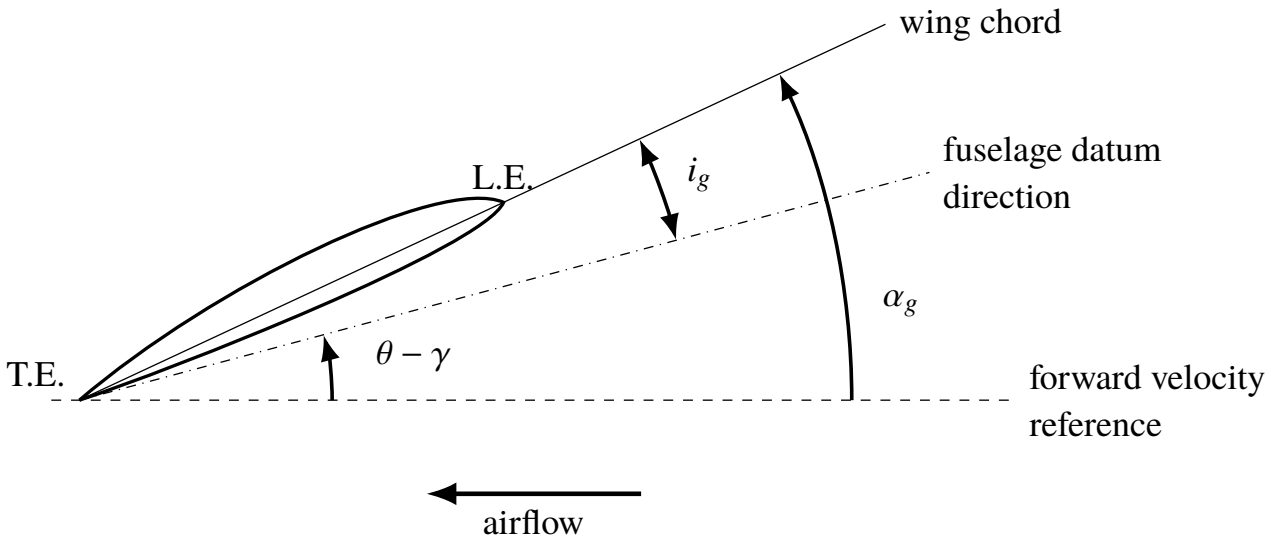
Can the nose be above the horizon while the aircraft descends? Identify the signs of θ and γ.

The fuselage is now located relative to the air. Next, include how the wing is mounted on that fuselage.

1 / AIRCRAFT ANGLES

Add the wing setting to find chord incidence

The wing chord joins the leading and trailing edges. Its fixed angle above the fuselage datum is the **chord setting** i_g.



The view is rotated so that the forward air-relative velocity reference is horizontal; the true horizon is omitted. The datum direction is translated parallel to itself for the angle construction.

Chord incidence α_g: the angle from the forward air-relative velocity direction to the chord.

Start at the flow reference, turn through $\theta - \gamma$ to reach the fuselage datum, then through i_g to reach the chord:

$$\alpha_g = (\theta - \gamma) + i_g = \theta + i_g - \gamma.$$

Read the signs physically: pitching up increases α_g; increasing the wing setting increases it; tilting the flight path upwards at fixed attitude decreases it.

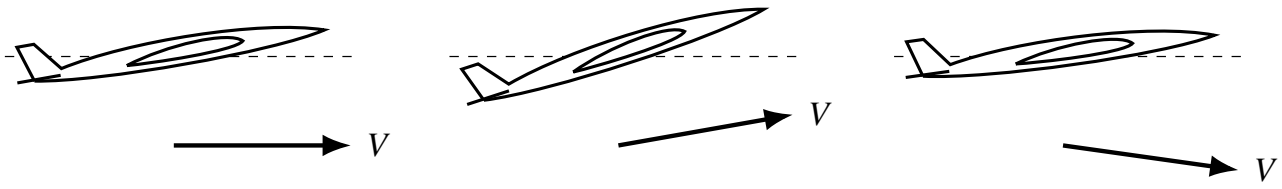
With the chord fixed in space, move the velocity direction closer to the chord. What happens to α_g?

This is a geometric angle. It can be calculated before choosing an aerodynamic lift model.

1 / AIRCRAFT ANGLES

Apply the geometry to level, climbing and descending paths

Use the chord-based relation $\alpha_g = \theta + i_g - \gamma$ in three prescribed flight geometries.



level
$\theta = 10°, \gamma = 0°$

climb
$\theta = 18°, \gamma = 10°$

descent
$\theta = 8°, \gamma = -8°$

In each case the chord setting is $i_g = 2°$. The schematic aircraft and path arrows show the signs; these are independent geometric examples.

Case	θ	γ	$\alpha_g = \theta + i_g - \gamma$
Level path	10°	0°	12°
Climbing path	18°	10°	10°
Descending path	8°	−8°	18°

Which case has the largest chord incidence? Why is it not the case with the highest nose attitude?

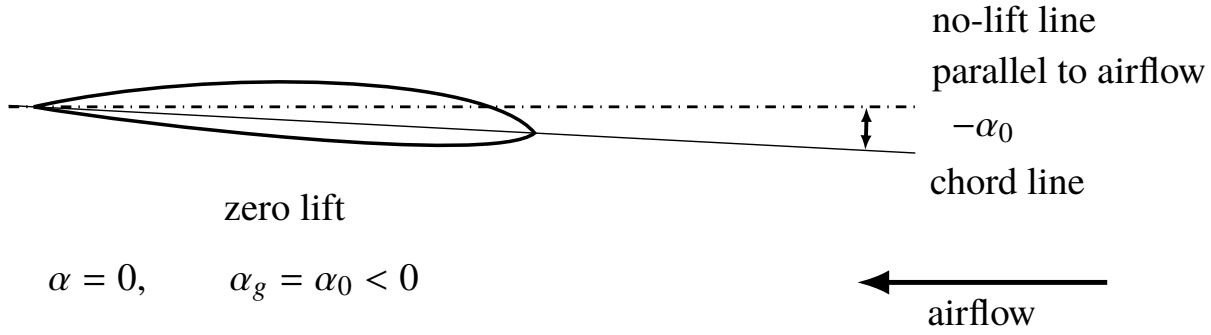
Attitude alone does not determine incidence. Whether a given incidence produces attached flow or stall depends on the wing's aerodynamic response.

1 / AIRCRAFT ANGLES

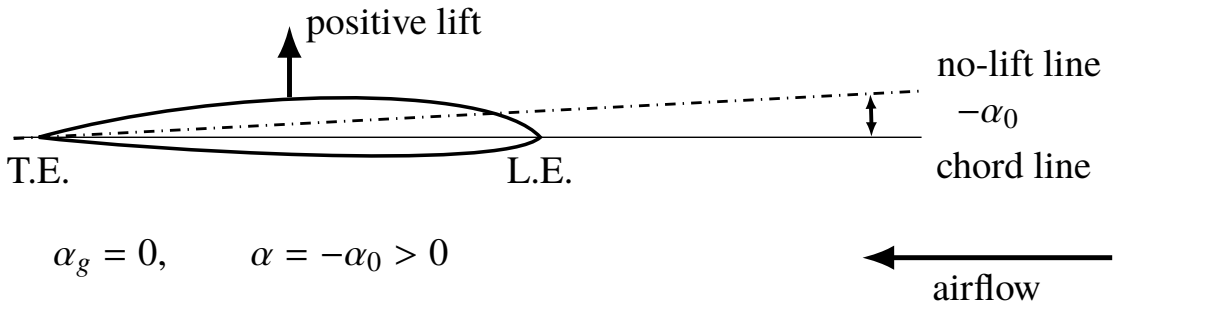
Find the aerofoil's zero-lift reference

The chord incidence is known. Does a chord aligned with the flow also mean zero lift? For a cambered aerofoil, generally it does not.

1. Rotate the section until its lift is zero



2. Rotate the same section until the chord is parallel to the airflow



The same cambered section is rotated in a fixed airflow. The unsigned gap between chord and no-lift directions is $-\alpha_0 = |\alpha_0|$ for $\alpha_0 < 0$.

Zero-lift angle α_0: the chord incidence at zero lift; it is negative for this positively cambered example.

No-lift line: a reference parallel to the airflow at zero lift. It rotates with the section; its angle to the chord is set by the zero-lift condition.

$$\alpha = \alpha_g - \alpha_0. \qquad \alpha_g = 0 \implies \alpha = -\alpha_0 > 0.$$

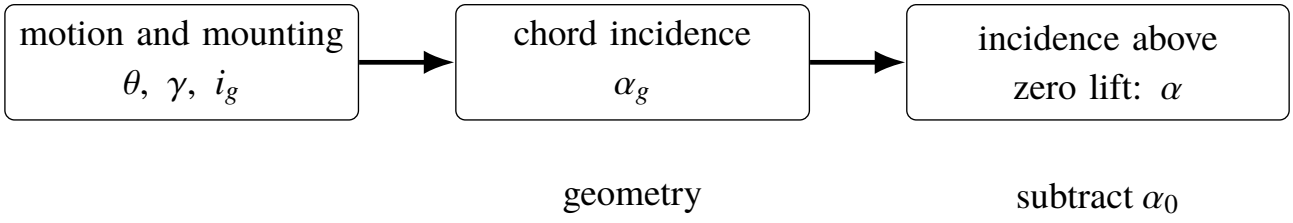
Model scope. In first-order thin-aerofoil theory, α_0 follows from the camber line; thickness does not contribute at that order. The reference stays fixed for a chosen configuration and flow condition in the approximately linear, attached-flow range. Real data can change with Reynolds number and separation.

Zero lift is not zero loading. Surface pressure, drag and a pitching moment can remain when the net lift is zero.

1 / AIRCRAFT ANGLES

Connect chord incidence to the zero-lift reference

The flight geometry has already given α_g. Camber adds a change of reference; it does not change the aircraft's orientation.



One forward calculation: first locate the chord relative to the air, then measure the same orientation from the section's zero-lift reference.

$$\alpha_g = \theta + i_g - \gamma, \qquad \alpha = \alpha_g - \alpha_0 = \theta + i_g - \gamma - \alpha_0.$$

For later wing–tail calculations it is convenient to define the matching no-lift-line setting:

$$i_w = i_g - \alpha_0 \quad \Longrightarrow \quad \boxed{\alpha = \theta + i_w - \gamma}.$$

Both versions give the same body-to-path angle:

$$\alpha_g - i_g = \alpha - i_w = \theta - \gamma.$$

Scope of the wing model. The simple untwisted wing model uses one representative chord and zero-lift reference. A twisted wing has spanwise-varying section references; a whole-wing zero-lift angle must come from the wing's integrated lift response.

Choose either (α_g, i_g) or (α, i_w) consistently.

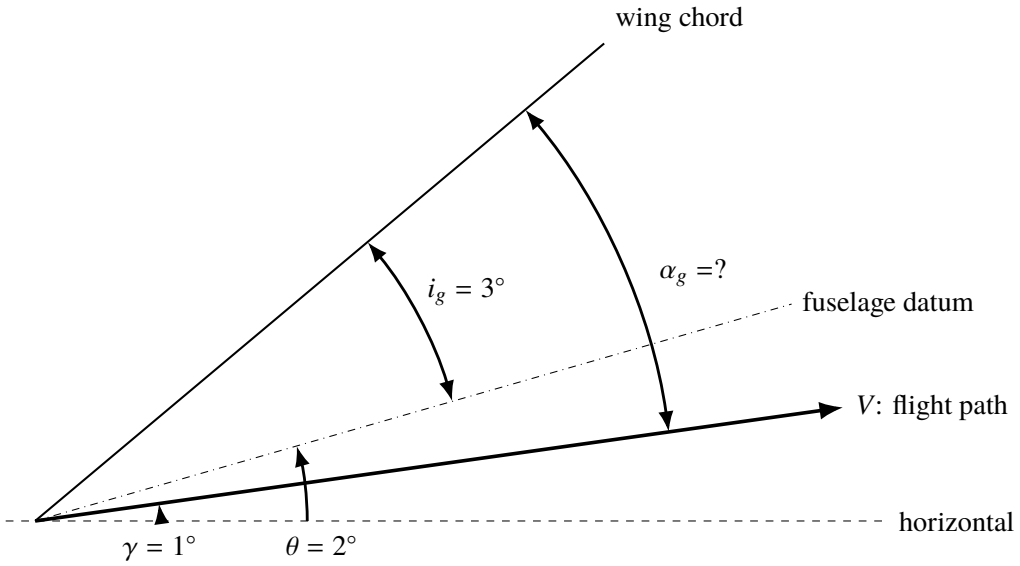
1 / AIRCRAFT ANGLES

Flight-test check: what incidence does the wing see?

During a flight test, an aircraft is making a shallow climb. At the instant being checked, its pitch attitude is $\theta = 2^\circ$ and its air-relative flight path is $\gamma = 1^\circ$ above horizontal.

The wing chord is mounted $i_g = 3^\circ$ above the fuselage datum. Aerodynamic data give a zero-lift angle $\alpha_0 = -1.2^\circ$ for the simple wing model.

Question. What wing incidence should be used to predict lift: first relative to the chord, then relative to zero lift?



Reference directions translated to one origin. Angular gaps are enlarged so they can be marked during the lecture; the labels give the flight-test values.

$$\text{chord above horizontal:} \quad \theta + i_g = 2 + 3 = 5^\circ,$$
$$\text{chord above flight path:} \quad \alpha_g = 5 - 1 = 4^\circ,$$
$$\text{incidence above zero lift:} \quad \alpha = 4 - (-1.2) = 5.2^\circ.$$

The fuselage is 2° nose-up, but the wing meets the air at 4° to its chord. Its lift law measured from zero lift uses 5.2°.

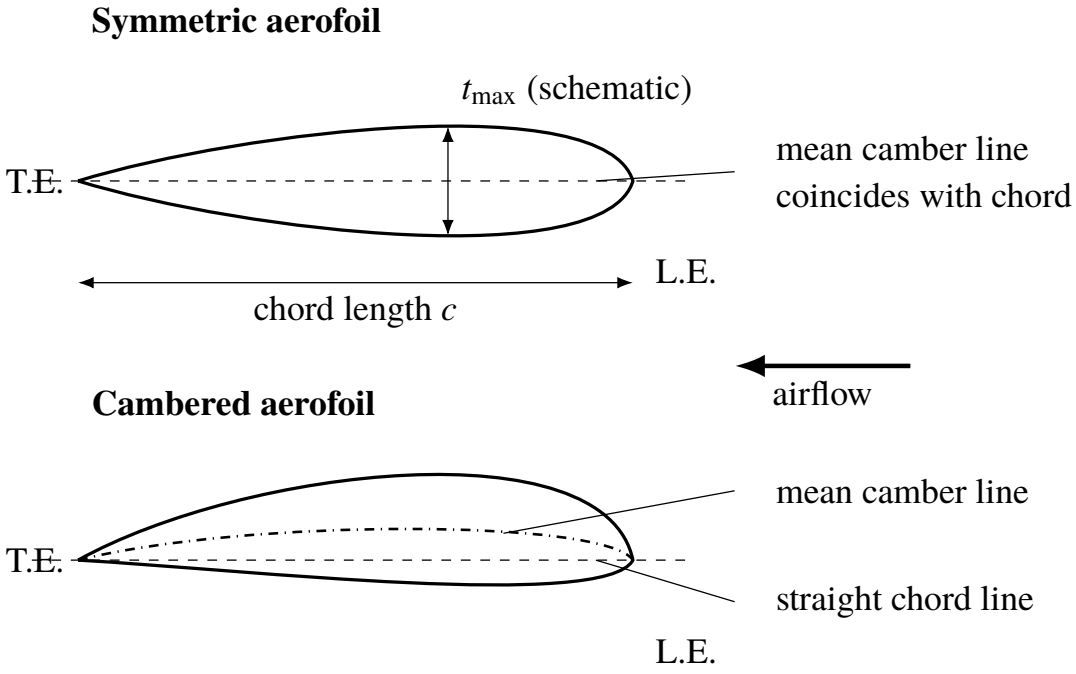
If the flight-path angle becomes 4° with attitude and setting unchanged, what are α_g and α?

The incidence needed for an aerodynamic calculation is now known. Next, establish how the aerofoil turns that incidence into lift and drag.

2 / AEROFOIL AERODYNAMICS - RECAP

Aerofoil aerodynamics: chord, thickness and camber

Aircraft motion and wing setting determine incidence. The next aim is to relate that incidence to lift and drag. Shape and incidence set the pressure distribution over the section; adding those surface forces gives its aerodynamic load.



- **Chord:** the straight line joining the leading and trailing edges; its length is c.
- **Mean camber line:** the line midway between the upper and lower surfaces. Its displacement from the chord describes camber.
- **Thickness:** separation of the upper and lower surfaces; t_{max}/c is the maximum thickness-to-chord ratio.

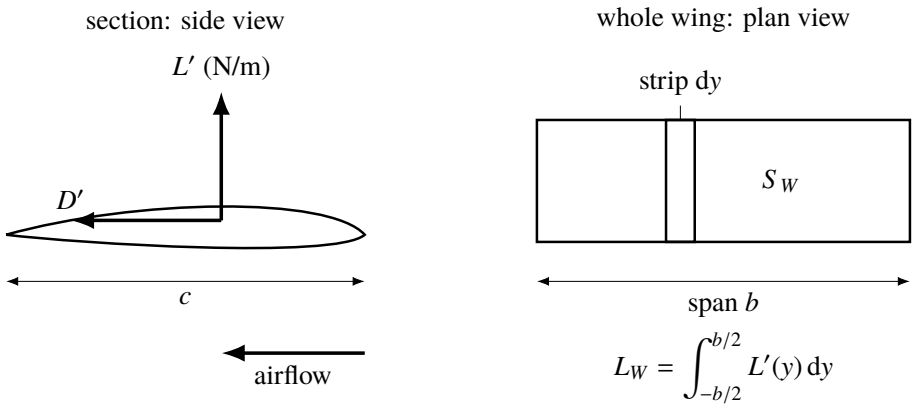
> For an ideal symmetric section, the chord is also the no-lift line. For a cambered section, the two reference directions generally differ.

Thickness and camber are different shape properties: a section can have finite thickness and zero camber.

2 / AEROFOIL AERODYNAMICS - RECAP

From an aerofoil section to lift and drag coefficients

An aerofoil is a two-dimensional section model. Its lift L' and drag D' are forces *per unit span*; a complete wing produces total forces L_W and D_W.



The section view shows the force directions. Adding the lift of the spanwise strips gives whole-wing lift: a load per metre becomes a total force.

The factor $\frac{1}{2}\rho V^2$ is the **dynamic pressure**.

	Aerofoil section	Whole wing
Lift	$c_l = \dfrac{L'}{\frac{1}{2}\rho V^2 c}$	$C_L^W = \dfrac{L_W}{\frac{1}{2}\rho V^2 S_W}$
Drag	$c_d = \dfrac{D'}{\frac{1}{2}\rho V^2 c}$	$C_D^W = \dfrac{D_W}{\frac{1}{2}\rho V^2 S_W}$
Load units	L', D' in N/m	L_W, D_W in N

Lower-case c_l, c_d describe a section; upper-case C_L^W, C_D^W describe the complete wing. V is the undisturbed speed.

Training-aircraft example: Diamond DA20-C1. Published wing area $S_W = 11.6\,\mathrm{m}^2$ and maximum mass 800 kg. For an illustrative condition, take $V = 40\,\mathrm{m/s}$, $\rho = 1.225\,\mathrm{kg/m}^3$ and $C_L^W = 0.70$:

$$\tfrac{1}{2}\rho V^2 = 980\,\mathrm{Pa}, \qquad L_W = 980(11.6)(0.70) = 7.96\,\mathrm{kN}.$$

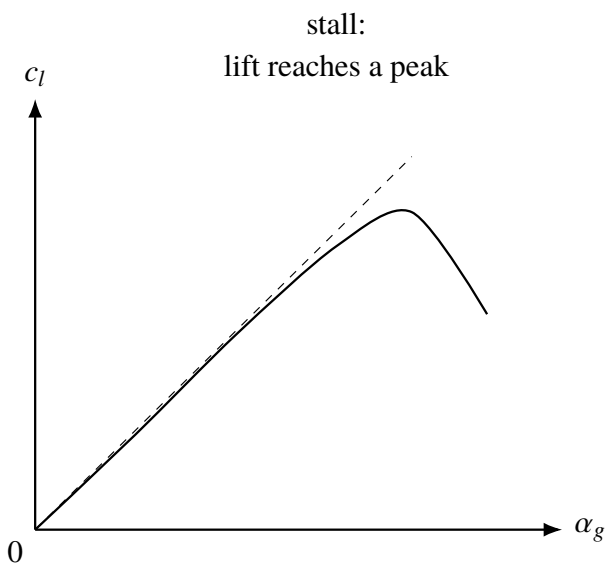
This is close to the aircraft's 7.85 kN weight at that mass: a realistic force scale for a light trainer. A hypothetical 1 m chord with $c_l = 0.70$ instead gives $L' = 686\,\mathrm{N/m}$, a local load per metre of span.

Aircraft data: EASA TCDS IM.A.223 (area); Diamond Aircraft training brochure (mass). Speed and lift coefficients are teaching assumptions.

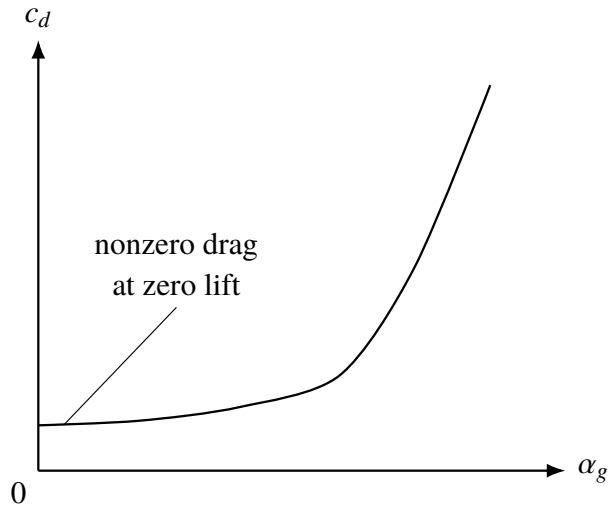
2 / AEROFOIL AERODYNAMICS - RECAP

A symmetric aerofoil: lift and drag behave differently

For an ideal symmetric section in symmetric flow, lift is zero when the chord is parallel to the airflow. The chord and no-lift line coincide, so $\alpha_0 = 0$ and $\alpha = \alpha_g$.



lift rises approximately linearly
before the stall region



drag is already present at zero lift
and rises strongly near stall

Qualitative section curves, shown from zero towards positive incidence. The vertical scales differ and no stall angle is prescribed. The dashed line is the attached-flow lift approximation.

$$c_l \simeq a_\infty \alpha \qquad \text{in the approximately linear range.}$$

Lift: the coefficient is nearly proportional to incidence until separation invalidates the straight-line model. The curve can depart from linearity before stall; the marked stall peak is the maximum lift coefficient in this sketch, not zero lift.

Drag: skin friction and the streamwise pressure force produce profile drag. Drag remains nonzero at zero lift and generally rises strongly as separation develops.

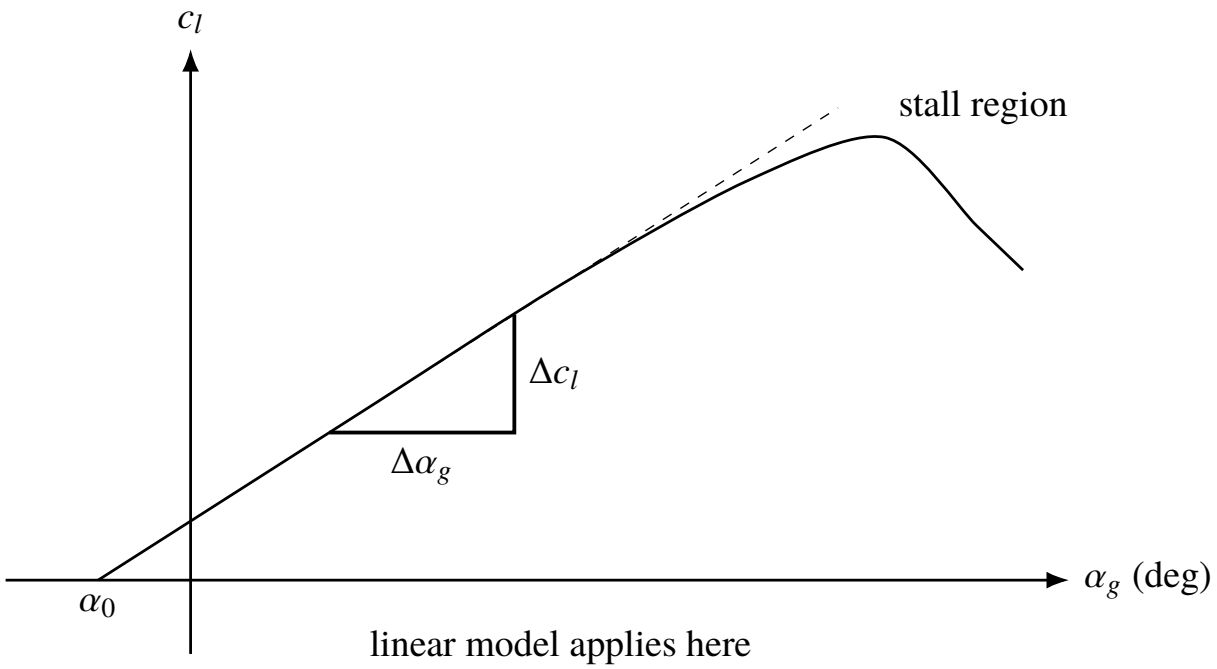
If $\alpha = 0$ and $c_l = 0$, must c_d also be zero? Locate that condition on both graphs.

The section drag shown here is profile drag. The extra induced drag caused by finite span enters later.

2 / AEROFOIL AERODYNAMICS - RECAP

A cambered aerofoil can lift at zero chord incidence

For the positively cambered section shown below, the lift curve crosses zero at a negative chord incidence α_0. The section therefore produces positive lift when its chord is parallel to the free stream.



Schematic cambered-section response. The straight part supports a linear model; extending it through the stall region does not.

$$c_l \simeq a_\infty(\alpha_g - \alpha_0) = a_\infty\alpha, \qquad \alpha = \alpha_g - \alpha_0.$$

Two reference lines describe the same section orientation: α_g uses the chord, and α uses the no-lift line.

At $\alpha_g = 0$**:** $c_l = -a_\infty\alpha_0 > 0$ if $\alpha_0 < 0$. **At** $\alpha_g = \alpha_0$**:** $c_l = 0$.

The lift law uses the angular distance from the zero-lift crossing to the operating point.

2 / AEROFOIL AERODYNAMICS - RECAP

Thin aerofoil theory: choose a slope and its units

The lift-curve slope gives the change in section lift coefficient for a small change in incidence:

$$\Delta c_l = a_\infty \Delta\alpha.$$

Thin aerofoil theory. For a thin two-dimensional section in incompressible, attached flow at small incidence, neglecting viscous effects,

$$a_\infty = 2\pi\,1/\text{rad} \simeq 6.283\,1/\text{rad}.$$

The ideal theory predicts the linear slope; it does not predict stall. Camber shifts the zero-lift angle in this first-order model.

Thickness-based slope estimate (Hoerner, *Lift*):

$$a_\infty \simeq 5.7\left(1 + 0.8\frac{t_{\max}}{c}\right) 1/\text{rad}.$$

For $t_{\max}/c = 0.12$, this gives $a_\infty \simeq 6.2472\,1/\text{rad}$. This is an alternative estimate of the section slope.

$$a_\infty = 2\pi\,1/\text{rad} \qquad \Delta\alpha = \pi/180\,\text{rad}$$

$$\longleftrightarrow$$

$$a_\infty = 0.1097\,1/\text{deg} \qquad \Delta\alpha = 1\,\text{deg}$$

$$\text{both give } \Delta c_l = 0.1097$$

The same one-degree incidence increase, using a slope per radian or per degree.

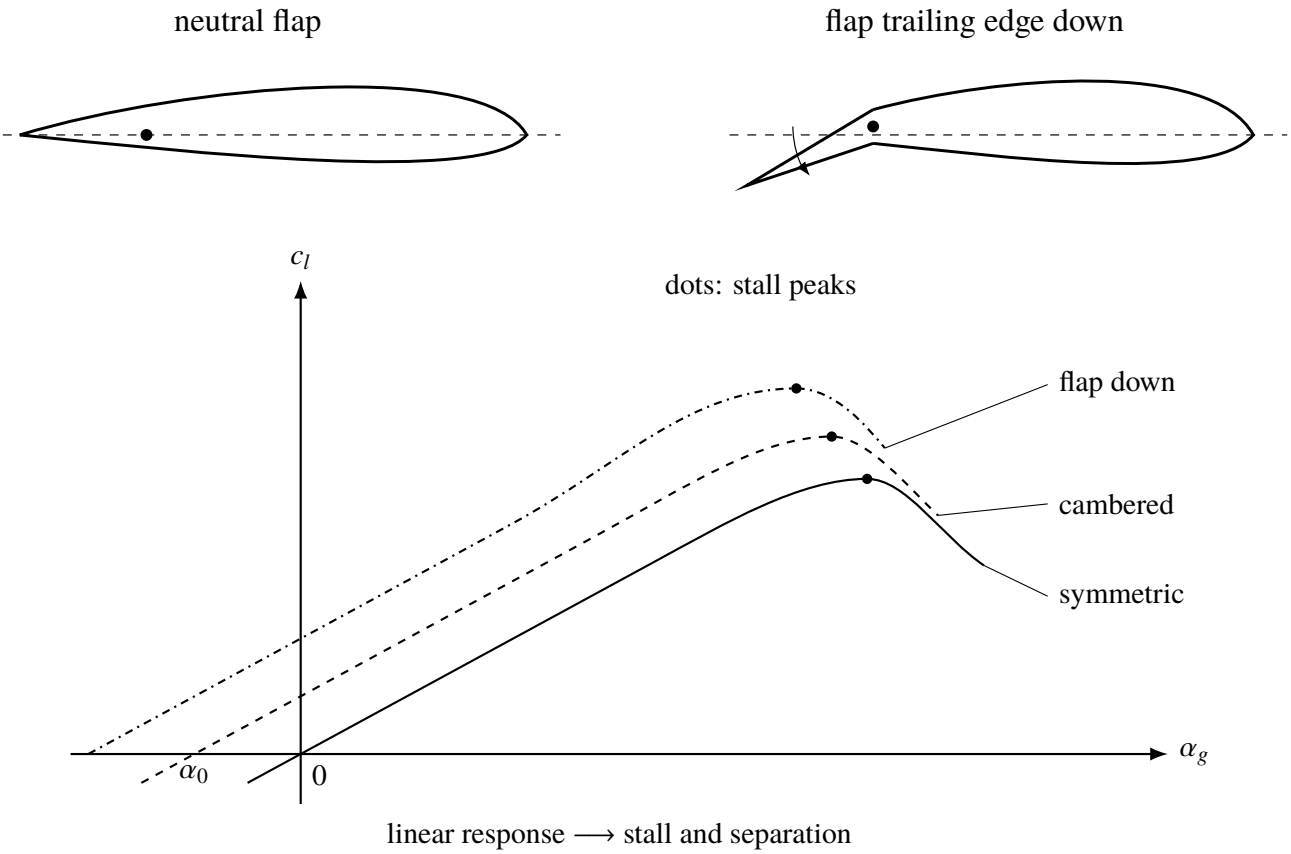
Unit conversion: $a_{\text{deg}} = a_{\text{rad}}\pi/180$. Use radians with a slope in $1/\text{rad}$.

Check: with $a_\infty = 2\pi\,1/\text{rad}$, a 2° increase gives $\Delta c_l = (2\pi)(2\pi/180) = 0.2193$.

2 / AEROFOIL AERODYNAMICS - RECAP

A flap changes camber and shifts the lift curve

A downward flap changes the section's effective camber. At the same chord incidence, it generally increases lift in the attached-flow range.



Qualitative curves on the same chord-incidence axis. Dots mark illustrative stall peaks; their angles and heights are not prescribed data. The nearly parallel portions compare the linear response only.

- Positive camber and flap deflection move the zero-lift crossing towards more negative incidence in this example.
- For the same lift coefficient in the linear range, less chord incidence is required. On approach, flaps also allow the required lift to be produced at a lower airspeed.
- The straight-line approximation fails before the stall peak. Flap deflection also changes maximum lift and stall behaviour.

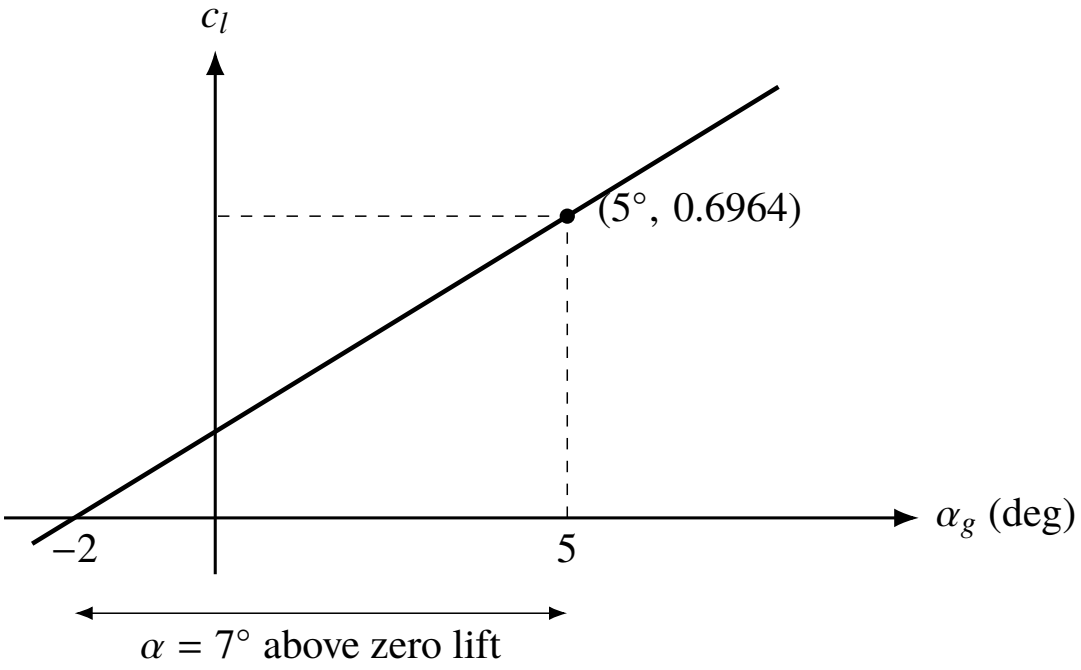
The control changes the lift response, but it does not remove the stall limit.

2 / AEROFOIL AERODYNAMICS - RECAP

Worked example: read a cambered aerofoil's lift curve

Given: $a_\infty = 5.7\,1/\text{rad}$, $\alpha_0 = -2°$, chord incidence $\alpha_g = 5°$ and a stated stall angle of 12°.

Find: incidence above zero lift and the section lift coefficient, using the supplied linear model.



The operating point is seven degrees above the zero-lift crossing. This is the given linear model; the graph does not predict the stalled response.

$$\alpha = \alpha_g - \alpha_0 = 5 - (-2) = 7° = 0.122173\,\text{rad},$$
$$c_l = a_\infty \alpha = 5.7(0.122173) = 0.6964.$$

The chord incidence is below the supplied stall angle. The calculation uses the example's stated linear model; being below stall does not make every real lift curve exactly straight.

> Using 5° directly would give $c_l = 0.4974$ and miss the camber contribution. First measure the incidence from the zero-lift condition, then convert it to radians.

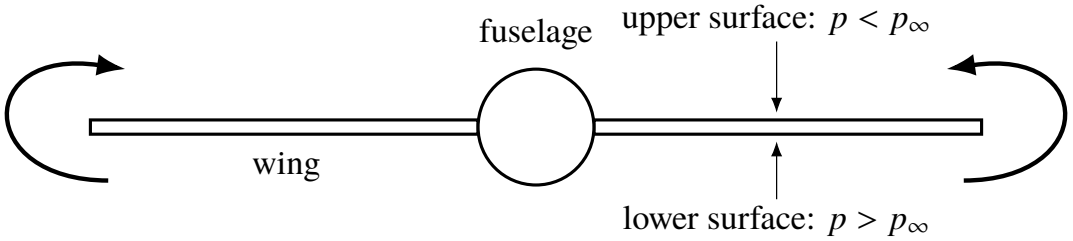
The section model is now established. A finite wing alters the local flow, so its lift slope will differ from the section slope.

3 / FROM SECTION TO WING

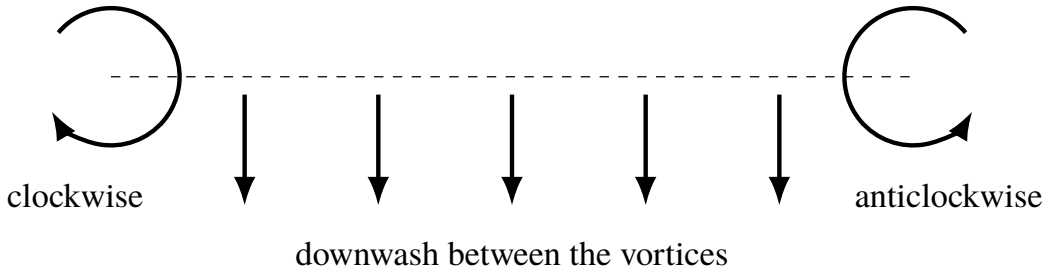
Finite span changes the airflow

An aerofoil section model has no wing tips. A real wing has a finite span, so air can move around its tips under the pressure difference between the lower and upper surfaces.

1. Pressure difference and flow around the tips



2. Trailing vortices induce downward motion



Rear-view schematics for positive lift. The upper drawing locates the pressure difference and fuselage; the lower drawing shows the trailing-vortex rotation and downward induced motion.

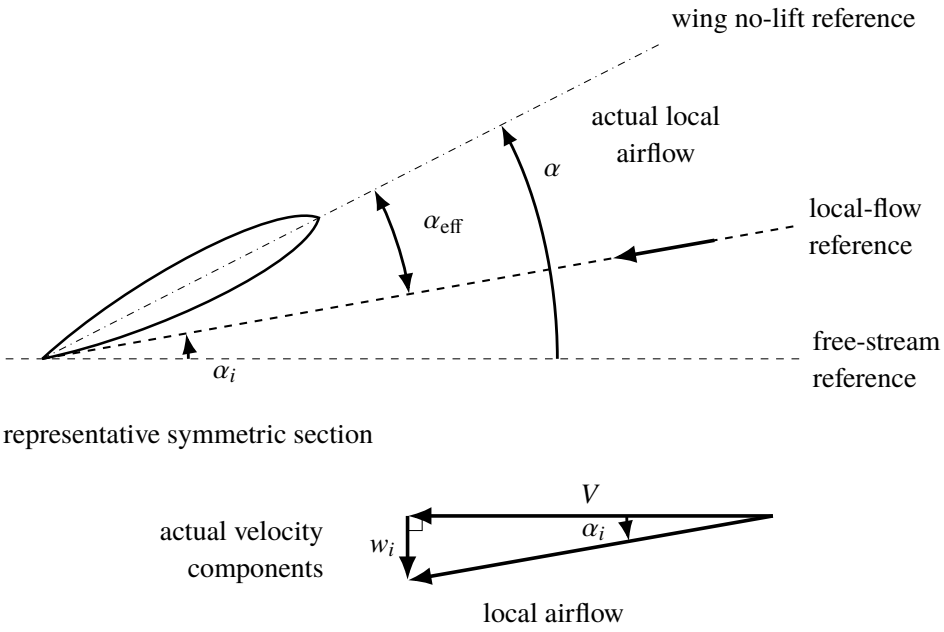
1. The pressure difference drives flow around the tips.
2. The wing's trailing vortex system induces a downward component of velocity at the wing.
3. The changed local-flow direction reduces effective incidence and tilts the aerodynamic force rearwards.

Compared at the same free-stream incidence: **a smaller lift response and an additional induced-drag component**.

The next steps are to draw the incidence reduction, derive the wing lift slope, and resolve the tilted force.

3 / FROM SECTION TO WING

The local section sees a smaller incidence



The local airflow travels backwards and downwards, so its forward reference tilts upwards towards the wing line.

$$\alpha_{\text{eff}} = \alpha - \alpha_i, \qquad \alpha_i \simeq \frac{w_i}{V}.$$

V is the undisturbed speed; w_i is the magnitude of the downward velocity induced at the wing. The velocity triangle gives $\tan\alpha_i = w_i/V$; at small angles, $\alpha_i \simeq w_i/V$ in radians.

The wing geometry enters through aspect ratio:

$$AR = \frac{b^2}{S_W}, \qquad b = \text{span}, \quad S_W = \text{wing area}.$$

For a rectangular wing this reduces to $AR = b/c$. A glider's long, slender wing is a familiar high-aspect-ratio example.

Ideal elliptically loaded, unswept wing:

$$\alpha_i = \frac{C_L^W}{\pi AR}.$$

Lift creates an induced angle, and that angle reduces the incidence producing lift. Both effects must be solved together.

3 / FROM SECTION TO WING

Derive the finite-wing lift slope

Model: unswept wing of sufficiently high aspect ratio, attached linear flow and ideal elliptic loading.

a_∞ is the section lift slope; a is the complete-wing slope. α is measured from zero lift using the undisturbed flow, and α_{eff} uses the local flow.

1. Apply the section response to the ideal wing:

$$C_L^W = a_\infty \alpha_{\text{eff}}.$$

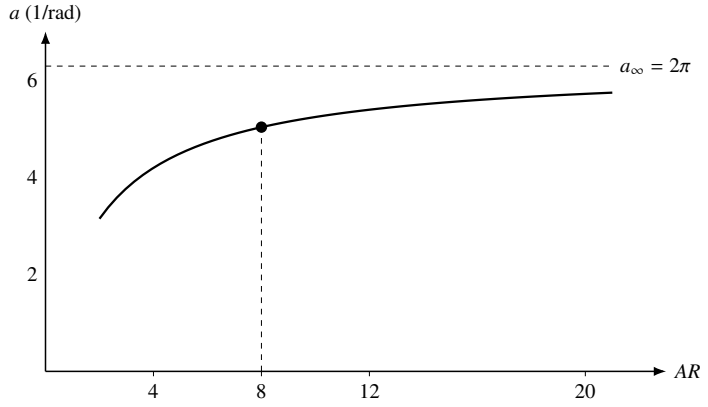
2. Substitute the induced-angle correction:

$$C_L^W = a_\infty \left(\alpha - \frac{C_L^W}{\pi AR} \right).$$

3. Collect the terms containing lift coefficient:

$$C_L^W \left(1 + \frac{a_\infty}{\pi AR} \right) = a_\infty \alpha.$$

$$C_L^W = a\alpha, \qquad a = \frac{a_\infty}{1 + a_\infty/(\pi AR)}.$$



For $a_\infty = 2\pi$ 1/rad, increasing aspect ratio brings the wing slope closer to the section slope.

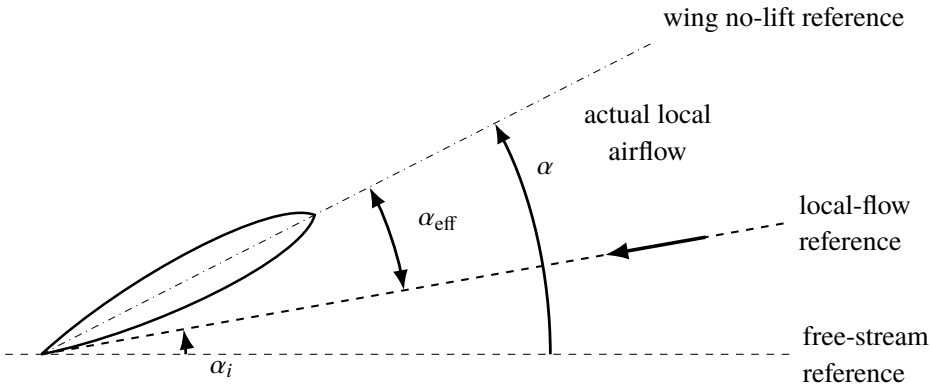
Why must $a < a_\infty$ in this model? What happens as AR becomes very large?

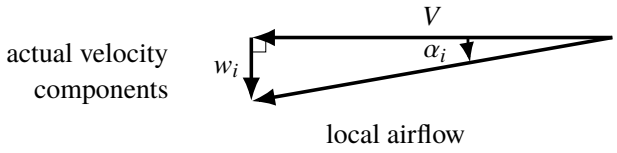
3 / FROM SECTION TO WING

Worked example: recover the effective angle

Given: $a_\infty = 2\pi\,1/\text{rad}$, $AR = 8$, $\alpha = 5°$; use the ideal model on page 19.

$$a = \frac{2\pi}{1 + 2/8} = 5.0265\,1/\text{rad},$$
$$C_L^W = 5.0265\left(5\frac{\pi}{180}\right) = 0.4386,$$
$$\alpha_i = \frac{0.4386}{\pi(8)} = 0.01745\,\text{rad} = 1.00°,$$
$$\alpha_{\text{eff}} = 5.00° - 1.00° = 4.00°.$$



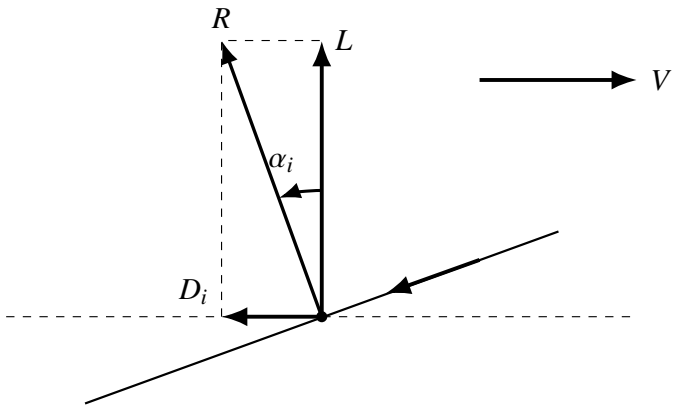


Read the answer back into the geometry: five degrees supplied, one degree lost to induced flow, four degrees effective.

Check: $a_\infty \alpha_{\text{eff}} = 2\pi(4\pi/180) = 0.4386$. Both routes give the same lift coefficient.

3 / FROM SECTION TO WING

Induced drag is a component of a tilted force



local airflow travels
backwards and downwards

The force normal to the local flow has an upward lift component and a rearward induced-drag component. Profile drag is omitted.

$$L = R\cos\alpha_i, \qquad D_i = R\sin\alpha_i.$$

$$\frac{D_i}{L} = \tan\alpha_i \simeq \alpha_i, \qquad D_i \simeq L\alpha_i.$$

The approximation uses radians.

One local-flow change	Two consequences
Smaller effective incidence	$\alpha_{\text{eff}} = \alpha - \alpha_i$
Rearward force component	$D_i/L \simeq \alpha_i$

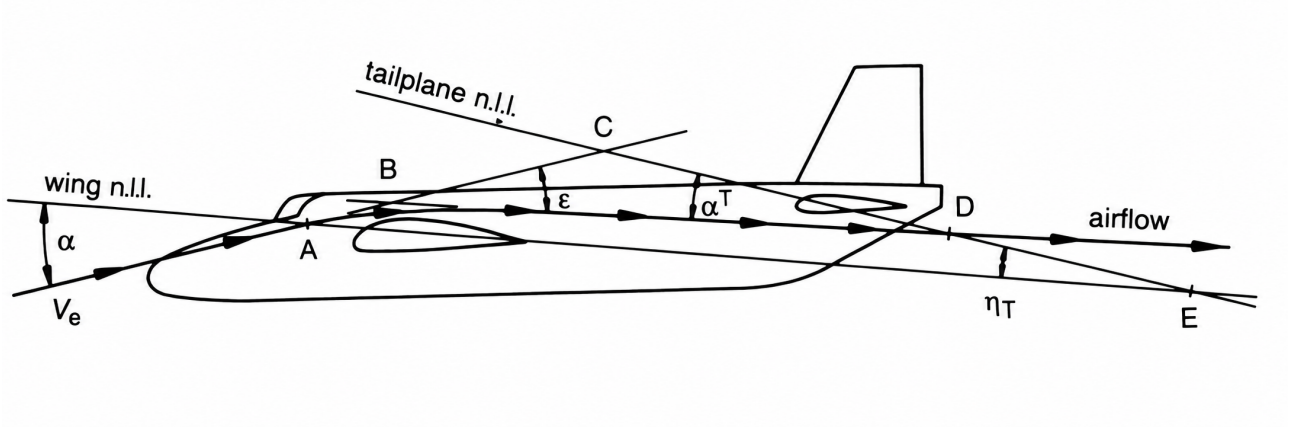
At fixed lift, what happens to induced drag if the small induced angle doubles?

Week 3 combines drag contributions in an aircraft polar. The next section follows the wing's disturbed flow downstream to the tail.

4 / THE TAILPLANE

The tail flies in the wing's disturbed flow

The wing changes the flow reaching the tail. Read the airflow arrows from left to right in this figure; the aircraft's nose is on the left.



Source: J. B. Russell, *Performance and Stability of Aircraft* (1996).

Read the lines first. AC is the straight extension of the undisturbed approach-flow direction; BD is the local flow at the tail. AE and CE are the wing and tailplane no-lift lines.

Angle	What it compares
α	Wing no-lift line AE with the undisturbed flow direction AC.
ε	Undisturbed direction BC with local tail flow BD: the downward turning of the flow.
α_T	Tailplane no-lift line CD with local flow BD.
η_T	Tailplane and wing no-lift lines, CE and AE: the fixed tail setting.

The figure's n.l.l. means *no-lift line*; its α^T is written α_T in the equations below. V_e labels the undisturbed approach flow; V denotes its speed in the equations.

> Downwash changes the flow direction at the tail. With aircraft attitude and tail setting fixed, more downwash reduces tail incidence. The angle ε at the tail need not equal α_i at the wing.

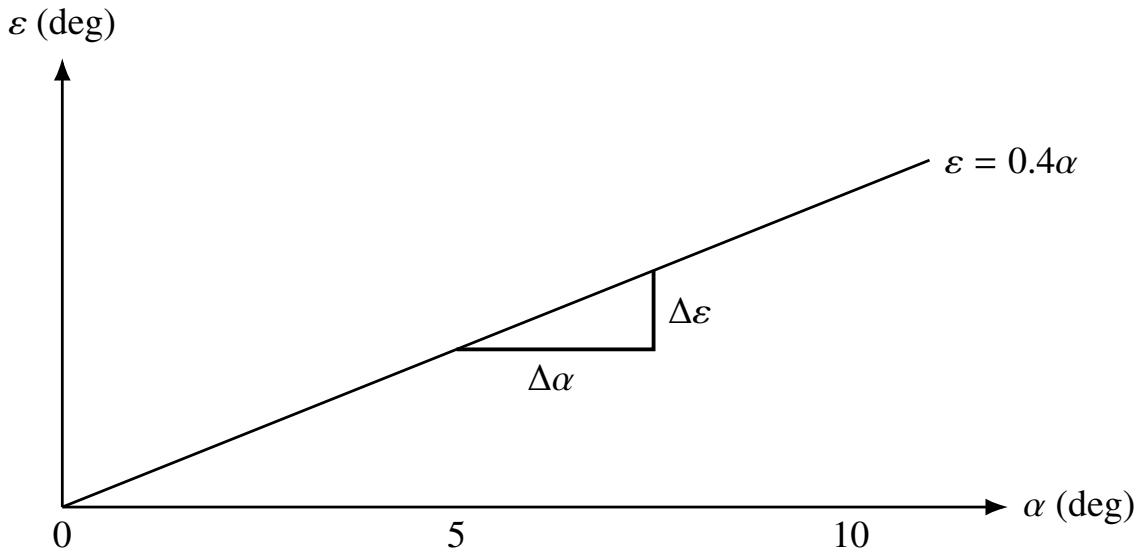
Here the local tail speed is approximated by V, so its dynamic pressure is $\frac{1}{2}\rho V^2$.

4 / THE TAILPLANE

A downwash gradient describes a response

For the fixed configuration used in these examples, approximate the downwash at the tail by a straight line through the origin:

$$\varepsilon = k_\varepsilon \alpha, \qquad k_\varepsilon = \frac{\mathrm{d}\varepsilon}{\mathrm{d}\alpha}.$$



The gradient measures the extra downwash produced by an incidence increment. This graph uses $k_\varepsilon = 0.4$.

$$\Delta\alpha = 1^\circ \quad \Longrightarrow \quad \Delta\varepsilon = 0.4^\circ.$$

Example: $\alpha = 6^\circ$ gives $\varepsilon = 2.4^\circ$. Both angles may be in degrees or both in radians; their ratio is the same.

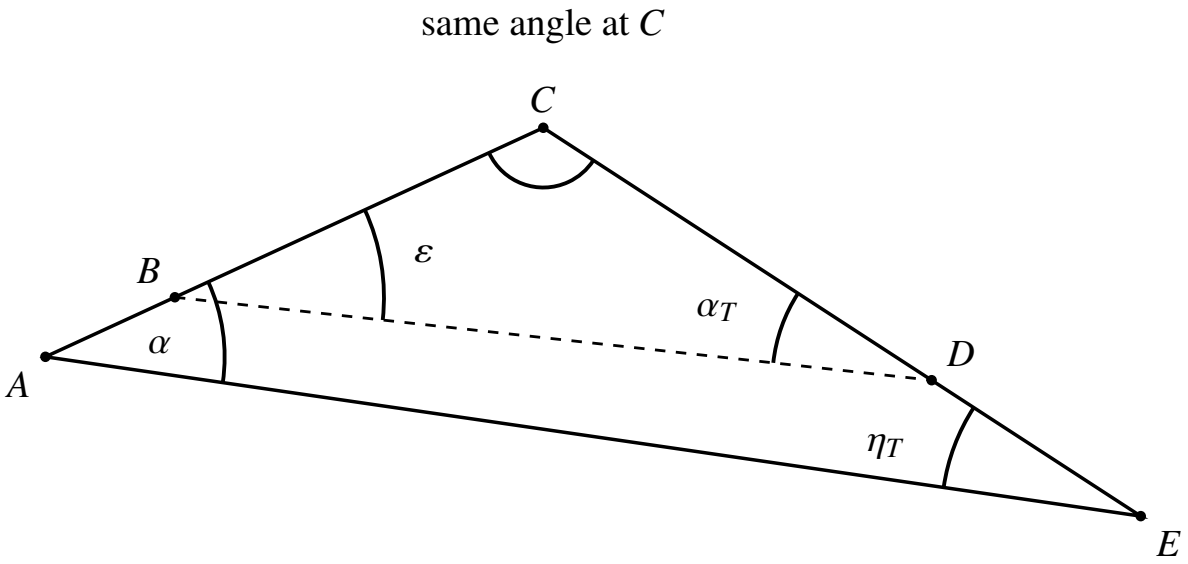
The input α uses the wing's no-lift reference. A different configuration or reference may require an offset; the through-origin model is the stated approximation for this lecture.

More wing incidence also means more downwash at the tail. The tail therefore receives only part of the incidence increase.

4 / THE TAILPLANE

Use the two triangles to find tail incidence

Use the construction on page 22. The two triangles ACE and BCD have the same angle at C, because A, B, C lie on one straight line and C, D, E lie on another.



The straight construction lines from the aircraft figure, with the body omitted. B lies on AC and D lies on CE; the dashed segment BD is the local-flow direction.

$$\begin{aligned} \text{triangle } ACE: &\qquad \alpha + \eta_T + \angle C = 180^\circ, \\ \text{triangle } BCD: &\qquad \varepsilon + \alpha_T + \angle C = 180^\circ. \end{aligned}$$

The common angle cancels, so the other two angles have the same sum:

$$\alpha_T + \varepsilon = \alpha + \eta_T \quad \Longrightarrow \quad \alpha_T = \alpha + \eta_T - \varepsilon.$$

The illustrated tail setting is positive. In calculations, keep η_T signed: positive means that the tail no-lift direction is nose-up relative to the wing's. The equation also applies to the negative settings used in these examples.

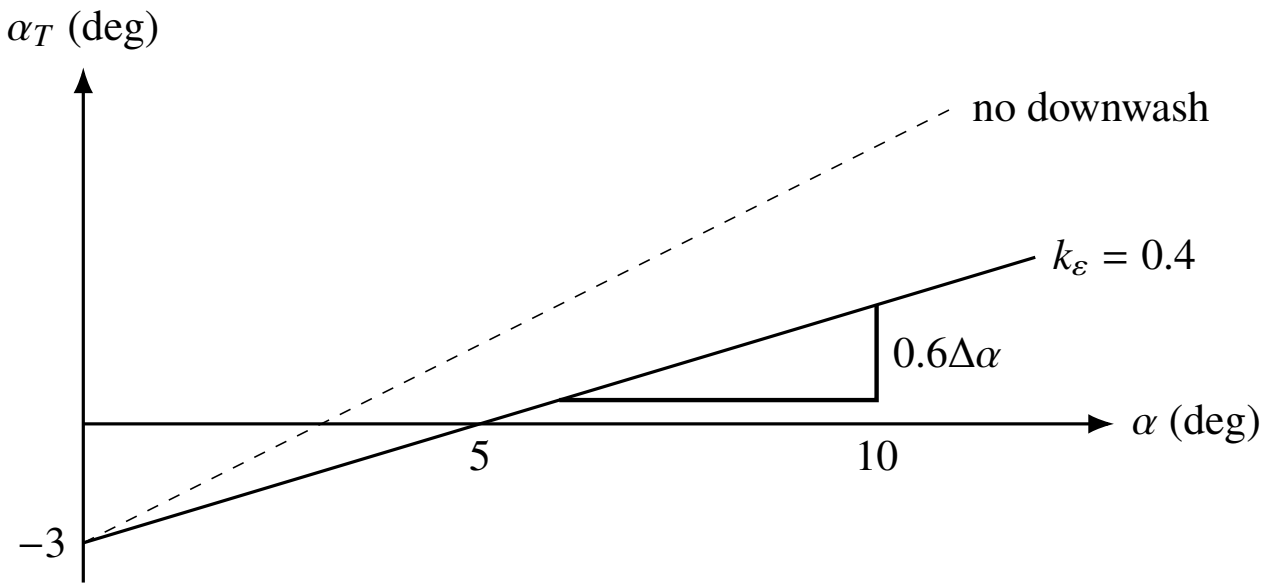
Hold α and η_T fixed. If the flow turns farther downwards, why must α_T decrease?

4 / THE TAILPLANE

The tail sees part of an incidence increment

Substitute the zero-offset downwash model into the geometry:

$$\alpha_T = (1 - k_\varepsilon)\alpha + \eta_T, \qquad \frac{\mathrm{d}\alpha_T}{\mathrm{d}\alpha} = 1 - k_\varepsilon.$$



For $\eta_T = -3^\circ$, the downwash gradient $k_\varepsilon = 0.4$ reduces the response slope from 1 to 0.6. The setting determines the intercept.

$$\Delta\alpha = 1^\circ \quad \Longrightarrow \quad \Delta\alpha_T = 0.6^\circ.$$

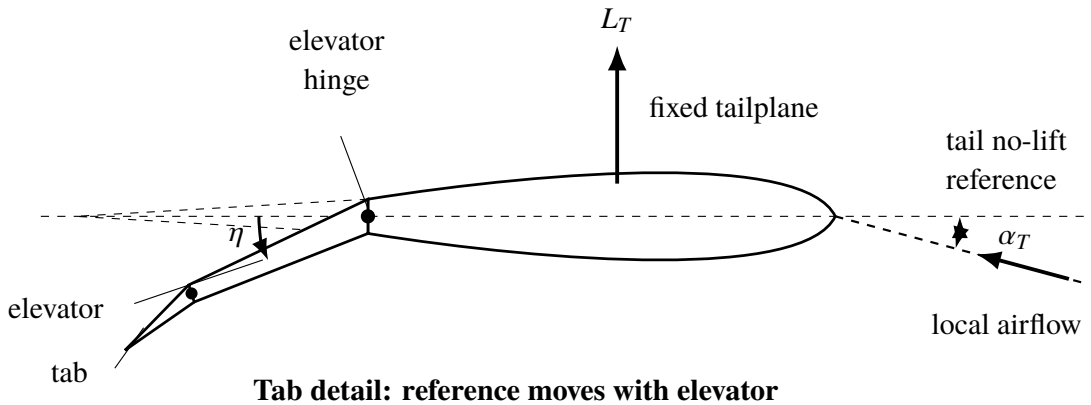
Does changing the fixed tail setting change the slope of these straight lines?

Setting shifts the line; the downwash gradient changes its slope. This distinction will matter in static stability.

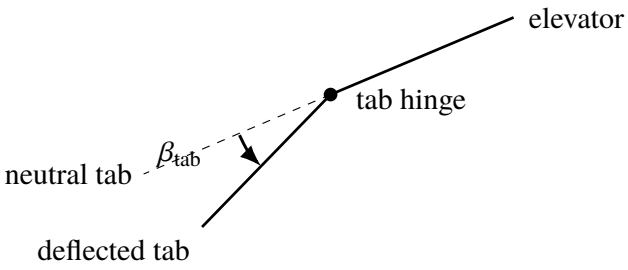
4 / THE TAILPLANE

Elevator and tab angles use their own references

The fixed tailplane, hinged elevator and smaller tab have separate roles. The dashed outlines below mark their neutral positions.



Tab detail: reference moves with elevator



Side view rotated to make the tail no-lift reference horizontal; leading edge to the right. Elevator angle η is measured from neutral; tab angle β_{tab} is relative to the elevator. Both are positive with their trailing edges down.

α_T	tail no-lift reference relative to local flow
η	elevator deflection from neutral
β_{tab}	tab deflection relative to the elevator

Moving the elevator changes the tail's aerodynamic loading. The tab can help balance the aerodynamic torque about the elevator hinge.

A tail force L_T and an elevator hinge torque H are different outputs. The following pages calculate each output separately.

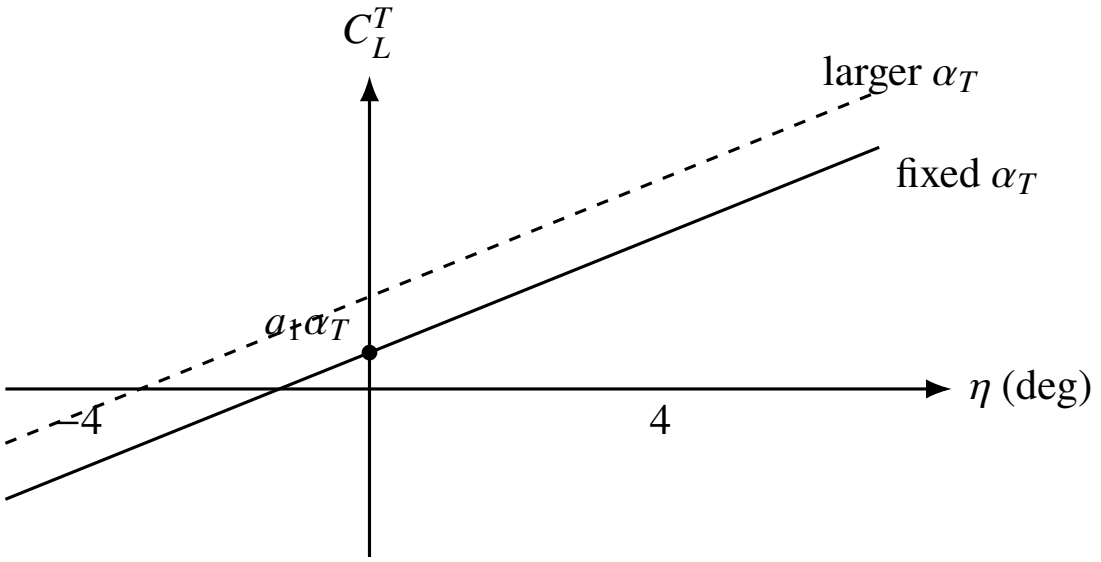
The subscript in β_{tab} distinguishes the tab angle from aircraft sideslip, introduced later.

4 / THE TAILPLANE

Add incidence and control contributions to tail lift

At a prescribed local incidence, add the contributions of the elevator and tab:

$$C_L^T = a_1\alpha_T + a_2\eta + a_3\beta_{\text{tab}}, \qquad L_T = \tfrac{1}{2}\rho V^2 S_T C_L^T.$$



At fixed incidence, elevator deflection changes tail lift along a straight line. Increasing incidence shifts the line upwards.

Coefficient	Response when the other inputs stay fixed
a_1	tail-incidence response
a_2	elevator response
a_3	tab response

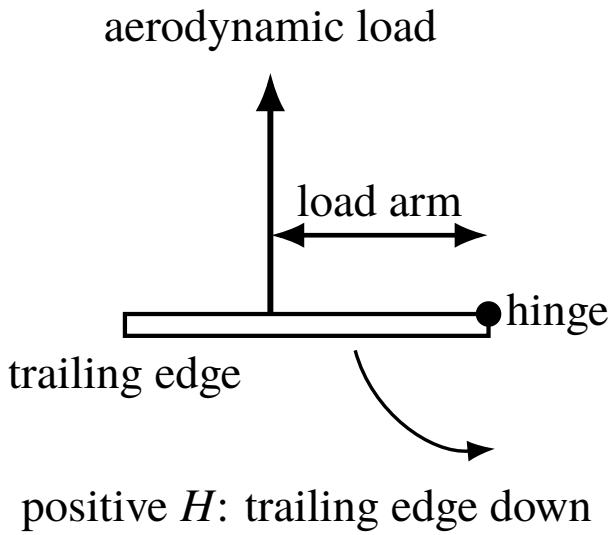
Use angles in radians with the supplied per-radian coefficients. All three contributions use tail area S_T.

The model takes zero lift at $\alpha_T = \eta = \beta_{\text{tab}} = 0$.

The first equation gives a coefficient; dynamic pressure and tail area turn it into a force.

4 / THE TAILPLANE

Hinge moment is a torque about the elevator hinge



upward load aft of hinge
tends to raise the trailing edge:
negative H in this sketch

The indicated elevator load turns about the hinge. It produces negative H under the positive trailing-edge-down convention.

$$C_H = b_1\alpha_T + b_2\eta + b_3\beta_{\text{tab}}, \qquad H = \tfrac{1}{2}\rho V^2 S_e \bar{c}_e C_H.$$

The inputs are the same as for tail lift, but the b coefficients describe **hinge torque**. Use the elevator reference area S_e and chord $\bar{c}_e$ supplied with those coefficients.

$$\underbrace{\tfrac{1}{2}\rho V^2 S_e}_{\text{force scale}}\ \underbrace{\bar{c}_e}_{\text{length scale}} \quad \longrightarrow \quad H \text{ in N m.}$$

For an elevator held stationary, the opposing holding torque is $H_{\text{hold}} = -H$. A tab can reduce hinge torque while the tail continues to carry lift.

Where does each quantity act: tail force L_T, aircraft pitching moment M, and elevator hinge torque H?

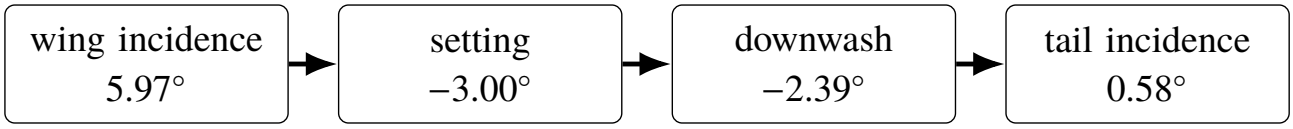
4 / THE TAILPLANE

Worked example: from wing incidence to tail load

Given: $C_L^W = 0.50$, wing slope $a = 4.8\,1/\text{rad}$, tail setting $\eta_T = -3°$, downwash $\varepsilon = 0.4\alpha$, and tail slope $a_1 = 3.2\,1/\text{rad}$. Elevator and tab are neutral.

$$\rho = 1.225\,\text{kg/m}^3, \quad V = 80\,\text{m/s}, \quad S_T = 10\,\text{m}^2.$$

Assume the speed at the tail equals V. **Find the tail lift.** First predict whether its incidence is larger or smaller than the wing's.



Use degrees to read the angle picture and unrounded radians for the lift calculation.

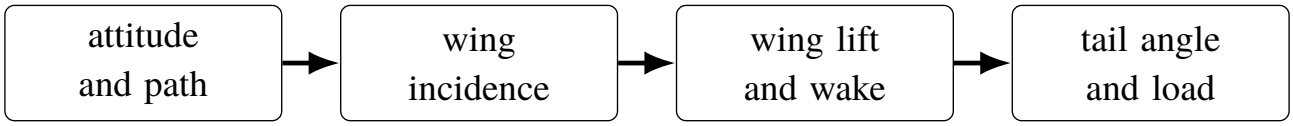
$$\begin{aligned}
\alpha &= \frac{0.50}{4.8} = 0.104167\,\text{rad},\\
\varepsilon &= 0.4\alpha = 0.041667\,\text{rad},\\
\alpha_T &= \alpha + \eta_T - \varepsilon = 0.010140\,\text{rad},\\
C_L^T &= a_1\alpha_T = 0.03245,\\
\tfrac{1}{2}\rho V^2 &= 3920\,\text{Pa}.
\end{aligned}$$

$$L_T = \tfrac{1}{2}\rho V^2 S_T C_L^T \simeq 3920(10)(0.03245) = 1272\,\text{N} \quad \text{upwards.}$$

The small positive tail incidence is about 0.58°. The negative setting and downwash have removed most of the wing's 5.97° incidence.

Conclusions

The complete calculation chain



The calculation now connects prescribed aircraft motion to wing and tail loads.

$$
\begin{aligned}
\text{wing incidence:} \quad & \alpha = \theta + i_w - \gamma, \\
\text{wing lift:} \quad & C_L^W = a\alpha, \\
\text{tail flow:} \quad & \varepsilon = k_\varepsilon \alpha, \\
\text{tail incidence:} \quad & \alpha_T = \alpha + \eta_T - \varepsilon, \\
\text{tail force:} \quad & L_T = \tfrac{1}{2}\rho V^2 S_T (a_1 \alpha_T + a_2 \eta + a_3 \beta_{\text{tab}}).
\end{aligned}
$$

If downwash increases at fixed wing incidence and control settings, which steps change, and what happens to tail lift?

Week 3: force and moment equilibrium determine the loads the aircraft needs. These relations give the required angles and settings.